Developing Mathematical Models to Describe First Class Stamp Costs Over Time

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MATH 340: Mathematical Modeling

Introduction

The cost of first-class postage stamps in the United States has changed significantly over time. In 1885, a postage stamp cost only $0.02, but by 2012, the price had increased to $0.45. These changes raise the question of whether historical stamp prices and mathematical models can be used to predict future costs. This is with consideration that the United States has known to have experienced moments of unpredictability and uncertainty where factors such as foreign wars, international conflicts, and inflation that has affected prices of everyday things to fluctuate. This is more relevant today with ongoing international conflicts and changes in government policies that have affected gasoline and food prices, and in time can potentially influence current and future prices of first-class postage stamps by changes in U.S Postal Services.

The purpose of this project is to analyze historical postage stamp prices using two mathematical models: quadratic and cubic least-squares models. Using the data provided in Section 4.4, Problem 5, we organized the postage rates information in Excel, calculated divided differences, and developed the two mathematical models to describe and predict postage costs from 1885 through 2027. This included using both models to predict the price of a postage stamp on January 1, 2027, and estimating when the price could reach $1. Finally, we compared both models using techniques like sum of squared errors and general observable trends to determine which model would be more reliable for predicting future stamp prices.

Data Analysis

We used the historical postage stamp prices provided in the textbook and the data set given. To make the calculations easier, we scaled the years using Equation X, where t was defined as the scaled year, that is the number of years since 1885.

$$t = Year - 1885 \text{ (X)}$$

We used all 39 data points provided in the spreadsheet, ranging from 1885 to 2026. We decided not to eliminate any data points because they represented the actual historical changes in postage stamp prices. Although there were some unusual changes, such as the temporary price increase during World War I and the price decrease in 2016, these are still part of the historical data. Keeping all the data points allowed us to analyze the overall trend without excluding any historical observations. In addition to this, there were no evident outliers as seen from the scatter plot in Figure X, where the data points were observed to fluctuate drastically from the overall trend.

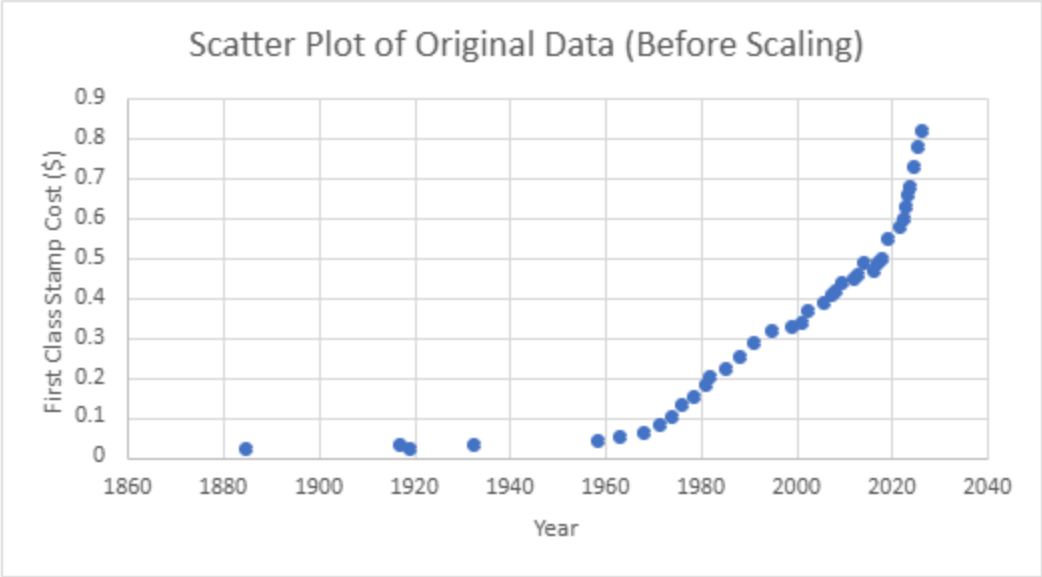


Figure X:

Divided Difference Table & Scatter Plot Behavior

To better examine how postage stamp prices have changed over time and what polynomial degree would be most accurate for modeling the historical postage data, a divided difference table was constructed up to the tenth order divided difference (Δ^{10}). Upon construction of the divided difference table (Appendix A), it was observed that the divided differences were not constant with positive and negative values fluctuating near zero. This was apparent with the second and third divided differences fluctuating near zero but not completely settling down. At higher order divided differences (Δ^{4} to Δ^{10}), the pattern remained consistent with values fluctuating between positive and negative values near zero. With the differences not stabilizing, the trends suggested that the data did not follow a simple polynomial pattern, at least one that could be confidently picked to model the data.

Therefore, with the divided differences results making it difficult to identify a clear polynomial order, the scatter plot behavior was next analyzed. From the upward curvature in the scatterplot (Figure X) and its similar resemblance to that of a quadratic or cubic polynomial, it was found that a quadratic or cubic model could be most appropriate for representing the overall trend of the data.

Mathematical Models

To analyze the historical postage stamp prices, we developed two least-squares models: a quadratic model and a cubic model. The least-squares method minimizes the sum of squared errors (SSE) between the actual stamp prices and the predicted values. For both models, we scaled the years using the t relation as shown previously by Equation X.

(a) Quadratic Least-Squares Model

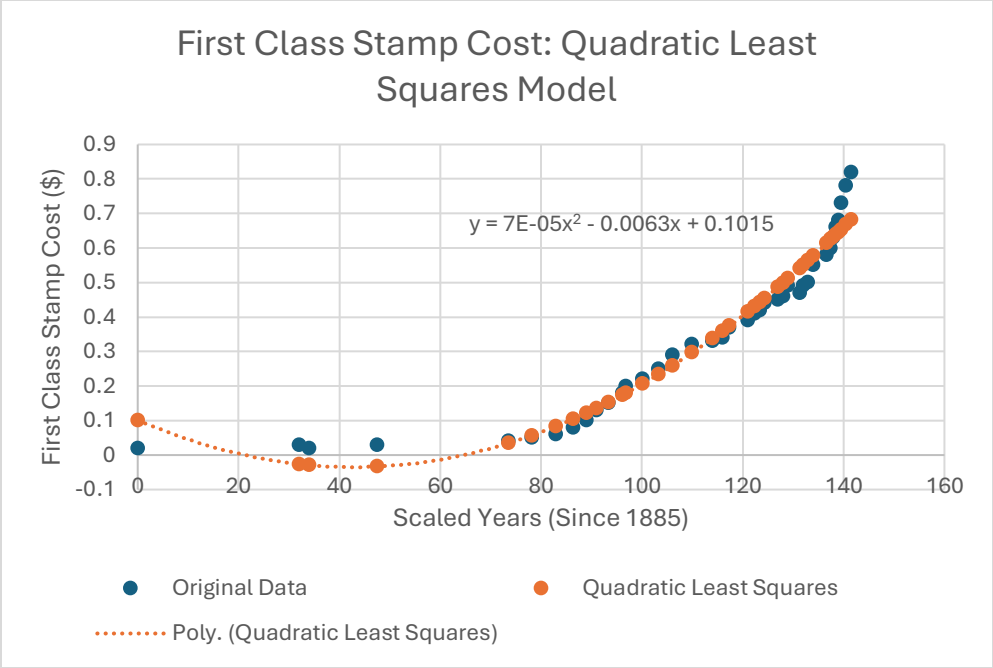


Figure X: Quadratic least-squares model compared with historical first-class postage stamp prices.

To calculate the quadratic least squares model, the general equation for a quadratic polynomial was first used (Equation X). To obtain the unknown coefficients c_1, c_2, and c_3 we first started by defining the sum of squared residuals as shown by Equation X. Then, the general second-degree polynomial was substituted into the least squares criterion (Equation X) to get Equation X:

$$P_2(t) = c_0 + c_1 t + c_2 t^2 \text{ (X)}$$

$$S = \sum_{i=1}^{m} \big(y_i - P_2(t)\big)^2 \text{ (X)}$$

$$S = \sum_{i=1}^{m} \Big(y_i - (c_0 + c_1 t_i + c_2 t_i^2)\Big)^2 \text{ (X)}$$

To minimize S, the partial derivatives with respect to the three unknown coefficients were taken and set equal to zero (Equation X) After differentiating and simplifying the resulting the three equations resulted from the partial derivatives, Equation X, X, and X were obtained. Subsequently, using Excel, the sums were calculated to obtain Equation X, X, and X.

$$\frac{\partial S}{\partial c_0} = 0, \ \frac{\partial S}{\partial c_1} = 0, \ \frac{\partial S}{\partial c_2} = 0 \text{ (X)}$$

$$m c_0 + \left(\sum_{i=1}^{m} t_i\right) c_1 + \left(\sum_{i=1}^{m} t_i^2\right) c_2 = \sum_{i=1}^{m} y_i \text{ (X)}$$

$$\left(\sum_{i=1}^{m} t_i\right) c_0 + \left(\sum_{i=1}^{m} t_i^2\right) c_1 + \left(\sum_{i=1}^{m} t_i^3\right) c_2 = \sum_{i=1}^{m} t_i y_i \text{ (X)}$$

$$\left(\sum_{i=1}^{m} t_i^2\right) c_0 + \left(\sum_{i=1}^{m} t_i^3\right) c_1 + \left(\sum_{i=1}^{m} t_i^4\right) c_2 = \sum_{i=1}^{m} t_i^2 y_i \text{ (X)}$$

$$39c_0 + 4184.8c_1 + 492329.4c_2 = 13.76$$

$$4194.8c_0 + 492336.4c_1 + 60107532.9c_2 = 1739.44$$

$$492326.4c_0 + 60107532.9c_1 + 7517604794.4c_2 = 223714.24$$

Lastly, to solve the system of equations and the three unknowns, the matrix form $A\vec{c} = \vec{y}$was applied. In this case, the matrix A was the coefficient matrix,

Using Excel, we calculated the coefficients and obtained:

(X)

The sum of squared errors for this model was:

$$SSE = 0.08029$$

Cubic Least-Squares Model

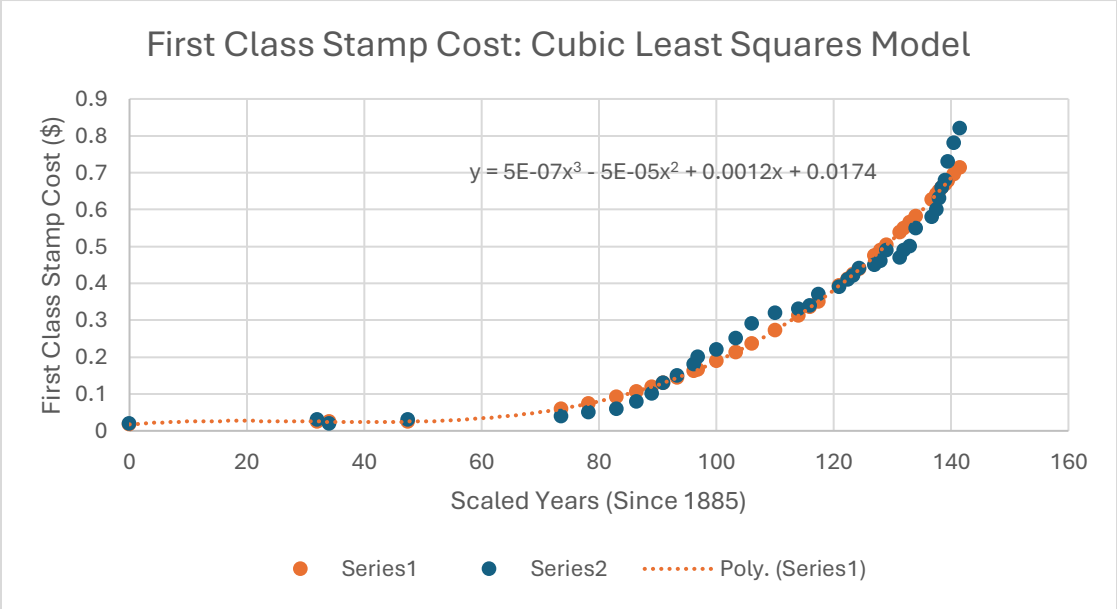


Figure X: Cubic least-squares model compared with historical first-class postage stamp prices.

The general equation for a cubic model is:

$$P_3(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$$

Using Excel, we calculated the coefficients and obtained:

$$P_3(t) = 0.0174 + 0.001186t - 0.0000459t^2 + 0.000000511t^3$$

The sum of squared errors for this model was:

$$SSE = 0.05362$$

Predictions

Using both the quadratic and cubic least-squares models, we predicted the cost of a first-class postage stamp on January 1st, 2027, and estimated when the price would reach $1.

Note: The equations shown in this report use rounded coefficients for simplicity. However, all predictions were calculated using the full-precision coefficients from Excel to maintain accuracy. As a result, substituting the rounded coefficients may produce slightly different answers.

a. Predicted Stamp Price on January 1, 2027

Since we scaled the years using $t = Year - 1885$, we first calculated:

$$t = 2027 - 1885$$

$$t = 142$$

i. Quadratic Model

Substituting $t = 142$ into our equation:

$$P_2(t) = 0.1015 - 0.00635t + 0.0000738t^2$$

$$P_2(142) \approx 0.6896$$

The quadratic model predicts that a postage stamp will cost approximately $0.69 on January 1, 2027.

ii. Cubic Model

Substituting $t = 142$ into our cubic equation:

$$P_3(t) = 0.0174 + 0.001186t - 0.0000459t^2 + 0.000000511t^3$$

$$P_3(142) \approx 0.7235$$

The cubic model predicts that a postage stamp will cost approximately $0.72 on January 1, 2027.

b. Predicting When a Stamp Will Reach $1

To determine when a postage stamp will reach $1, we set each model equal to 1 and solved for t.

i. Quadratic Model

$$1 = 0.1015 - 0.00635t + 0.0000738t^2$$

Solving using the precise coefficients:

$$t \approx 161.33$$

Converting back to the calendar year:

$$1885 + 161.33 = 2046.33$$

The quadratic model predicts that postage stamps will reach $1 around 2046

ii. Cubic Model

$$P_3(t) = 0.0174 + 0.001186t - 0.0000459t^2 + 0.000000511t^3$$

Solving using the precise coefficients:

$$t \approx 154.93$$

Converting back to the calendar year:

$$1885 + 154.93 = 2039.93$$

The cubic model predicts that postage stamps will reach $1 around 2040

c. Summary of Predictions

Prediction	Quadratic	Cubic
Stamp Price on January 1, 2027	$0.69	$0.72
Year stamp reaches $1	2046	2040

The cubic model predicts a slightly higher stamp price in 2027 and estimates that stamps will reach $1 approximately 6 years earlier than the quadratic model

Model Comparison

The cubic model had a smaller SSE of 0.05362 compared to the quadratic model's 0.08029, indicating a better fit for the historical data. This can also be seen in the graphs, where the cubic model follows the data more closely, especially during the earlier years. In contrast, the quadratic model predicts negative stamp prices during part of that period, which is unrealistic.

We believe the cubic model is the better predictor because it provides a more accurate fit for the historical data. However, its higher-degree term may cause prices to increase unrealistically over time, making its long-term predictions less reliable. Therefore, while the cubic model performs better overall, its future predictions should still be interpreted with caution.

Both models also predict a lower price for 2027 than the most recent observed price of $0.82 in 2026, suggesting that neither model fully captures the recent increases in stamp prices.

Conclusion

In this project, we analyzed historical postage stamp prices using quadratic and cubic least-squares models to predict future costs. Both models showed an overall increase in stamp prices over time, but the cubic model provided a better fit based on its smaller sum of squared errors.

Our results showed that the quadratic model predicted a stamp price of $0.69 in 2027 and $1 by 2046, while the cubic model predicted $0.72 in 2027 and $1 by 2040. Although we believe the cubic model is the better predictor, both models have limitations because future postage stamp prices depend on factors beyond historical trends.

Overall, this project demonstrated how mathematical modeling can be used to analyze historical data and make predictions while also showing the importance of considering the limitations of those predictions.

Appendix A: Divided Difference Table

Year	t= Year-1885	First Class Stamp Cost	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6	Δ^7	Δ^8	Δ^9	Δ^{10}
1885	0	0.02	0.00031 25	**-0.00015 625**	**1.10867 E-05**	-2.74551 E-07	6.4571E -09	-1.48803 E-10	7.31743 E-12	-6.28053 E-13	1.05187 E-13	-2.41354 E-14
1917	32	0.03	-0.005	**0.00037 037**	**-9.12021 E-06**	2.30394 E-07	-5.89353 E-09	4.83424 E-10	-4.85793 E-11	8.94396 E-12	-2.14906 E-12	3.50485 E-13
1919	34	0.02	0.00074 0741	**-9.03028 E-06**	**1.52402 E-06**	-7.01754 E-08	2.04047 E-08	-2.28559 E-09	4.79115 E-10	-1.23008 E-10	2.03521 E-11	-2.53838 E-12
1932.5	47.5	0.03	0.00038 3142	**5.83313 E-05**	**-1.91458 E-06**	9.99032 E-07	-1.05303 E-07	2.50239 E-08	-6.82757 E-09	1.14289 E-09	-1.39312 E-10	6.99138 E-12
1958.6	73.6	0.04	0.00217 3913	**-9.63614 E-06**	**3.69478 E-05**	-3.37104 E-06	9.83238 E-07	-2.88361 E-07	4.88312 E-08	-5.73913 E-09	2.28434 E-10	2.48439 E-11
1963.2	78.2	0.05	0.00208 3333	**0.00046 3295**	**-1.49663 E-05**	1.37373 E-05	-4.72632 E-06	8.15224 E-07	-8.48905 E-08	3.14386 E-10	9.66299 E-10	-1.49808 E-10
1968	83	0.06	0.00588 2353	**0.00030 1659**	**0.00016 0871**	-5.81027 E-05	9.94771 E-06	-7.72228 E-07	7.80054 E-08	-2.45685 E-08	3.21335 E-09	-2.63877 E-10
1971.4	86.4	0.08	0.00769 2308	**0.00158 8629**	**-0.00044 3397**	7.3207E -05	-7.8626E -07	2.10612 E-06	-4.20735 E-07	4.96599 E-08	-3.91134 E-09	2.3114E -10
1974	89	0.1	0.015	**-0.00151 5152**	**0.00027 4032**	-6.49513 E-05	2.96401 E-05	-5.0043E -06	5.57565 E-07	-4.26477 E-08	2.46813 E-09	-1.26361 E-10
1976	91	0.13	0.00833 3333	**0.00045 7875**	**-0.00078 7147**	0.00026 4054	-4.19214 E-05	4.53006 E-06	-3.38037 E-07	1.90556 E-08	-9.43616 E-10	4.19298 E-11
1978.4	93.4	0.15	0.01071 4286	**0.00510 2041**	**-0.00161 5744**	-0.00025 1579	2.64824 E-05	-1.89264 E-06	1.00243 E-07	-4.53478 E-09	1.63331 E-10	-9.10014 E-13
1981.2	96.2	0.18	0.02857 1429	**-0.00572 3443**	**0.00087 4891**	-8.47477 E-05	4.93546 E-06	-1.72355 E-07	2.24322 E-09	-6.14831 E-10	1.38215 E-10	-2.58404 E-11
1981.9	96.9	0.2	0.00625	**0.00048 8281**	**3.58882 E-05**	-1.66384 E-05	1.86754 E-06	-1.27939 E-07	1.52776 E-08	-2.81289 E-09	5.38804 E-10	-9.22059 E-11
1985.1	100.1	0.22	0.00937 5	**0.00081 8452**	**-0.00018 2075**	1.52964 E-05	-5.76107 E-07	1.85252 E-07	-5.2513E -08	1.09266 E-08	-1.90465 E-09	3.14204 E-10
1988.3	103.3	0.25	0.01428 5714	**-0.00098 4091**	**3.05455 E-05**	6.13633 E-06	2.62876 E-06	-9.1227E -07	1.9115E -07	-3.34518 E-08	5.73051 E-09	-7.92587 E-10
1991.1	106.1	0.29	0.00769 2308	**-0.00065 7254**	**0.00010 8477**	4.32018 E-05	-1.35184 E-05	2.7387E -06	-4.81231 E-07	8.7462E -08	-1.30538 E-08	2.09097 E-09
1995	110	0.32	0.0025	**0.00041 6667**	**0.00059 6658**	-0.00015 8223	3.11224 E-05	-5.5866E -06	1.11932 E-06	-1.85363 E-07	3.27385 E-08	-5.52686 E-09
1999	114	0.33	0.005	**0.00483 1933**	**-0.00114 3791**	0.00022 7695	-4.3738E -05	1.05317 E-05	-2.03184 E-06	4.0393E -07	-7.22719 E-08	8.86853 E-09
2001	116	0.34	0.02142 8571	**-0.00317 4603**	**0.00076 8849**	-0.00018 3442	6.57911 E-05	-1.58823 E-05	3.62318 E-06	-6.80149 E-07	8.11537 E-08	-5.59358 E-09
2002.4	117.4	0.37	0.00555 5556	**0.00174 6032**	**-0.00058 8624**	0.00036 9203	-0.00010 8914	2.75958 E-05	-5.21876 E-06	5.61502 E-07	-8.34367 E-09	1.62158 E-08
2006	121	0.39	0.01428 5714	**-0.00178 5714**	**0.00199 5798**	-0.00067 6371	0.00018 3602	-3.29418 E-05	2.58612 E-06	-4.39685 E-07	2.61311 E-07	-7.63917 E-08
2007.4	122.4	0.41	0.01	**0.005**	**-0.00206 243**	0.00060 8842	-7.99321 E-05	6.30468 E-06	-7.42265 E-06	2.69604 E-06	-7.31781 E-07	1.19159 E-07

2008.4	123.4	0.42	0.02	**-0.00447179**	**0.00134 7083**	8.12895 E-05	-0.00013 6044	6.49528 E-05	-2.11554 E-05	5.79261 E-06	-9.72192 E-07	1.54529 E-07
2009.4	124.4	0.44	0.00384 6154	**0.00170 9402**	**0.00180 2304**	-0.00099 3457	0.00042 255	-0.00013 8139	4.02463 E-05	-7.13754 E-06	1.20667 E-06	-2.02042 E-07
2012	127	0.45	0.01	**0.01**	**-0.00505 2548**	0.00221 7925	-0.00076 5446	0.00024 8225	-4.75455 E-05	8.6698E -06	-1.54111 E-06	2.42152 E-07
2013	128	0.46	0.03	**-0.01172 5955**	**0.00603 7079**	-0.00237 4752	0.00097 213	-0.00021 2966	4.34874 E-05	-8.28238 E-06	1.24365 E-06	-8.90682 E-08
2014	129	0.49	-0.00869 5652	**0.01242 236**	**-0.00583 6682**	0.00345 8031	-0.00088 0674	0.00020 0164	-3.93365 E-05	4.77589 E-06	2.63896 E-07	1.70809 E-07
2016.3	131.3	0.47	0.02857 1429	**-0.01092 437**	**0.01145 347**	-0.00332 3161	0.00082 0718	-0.00015 3865	6.0345E -06	7.41485 E-06	2.05739 E-06	-3.94929 E-06
2017	132	0.49	0.01	**0.02**	**-0.00649 1598**	0.00176 5291	-0.00021 0175	-0.00011 0416	6.31288 E-05	2.42855 E-05	-3.4276E -05	1.66453 E-05
2018	133	0.5	0.05	**-0.01051 0511**	**0.00321 7503**	0.00050 4241	-0.00092 7881	0.00033 1486	0.00024 527	-0.00026 7061	0.00012 3855	
2019	134	0.55	0.01111 1111	**0.00396 8254**	**0.00573 8706**	-0.00459 9105	0.00106 1032	0.00192 5739	-0.00175 7686	0.00078 5703		
2021.7	136.7	0.58	0.025	**0.02692 3077**	**-0.01495 7265**	0.00070 6057	0.01165 2599	-0.00949 9222	0.00413 5089			
2022.5	137.5	0.6	0.06	**0**	**-0.01333 3333**	0.03333 3333	-0.02444 4444	0.01034 9206				
2023	138	0.63	0.06	**-0.02**	**0.05333 3333**	-0.04	0.01695 2381					
2023.5	138.5	0.66	0.04	**0.06**	**-0.04666 6667**	0.01933 3333						
2024	139	0.68	0.1	**-0.03333 3333**	**0.01133 3333**							
2024.5	139.5	0.73	0.05	**-0.005**								
2025.5	140.5	0.78	0.04									
2026.5	141.5	0.82										